

Dynamic Perception: How Prediction, Attention, and Neural Plasticity Shape What We Perceive

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Executive Summary

Perception is not a passive reception of sensory reality. Rather, it is a fundamentally dynamic process in which the brain actively generates predictions about the world, filters these predictions through selective attention, and continuously refines them through experience-dependent neural plasticity. This chapter synthesizes three seemingly distinct neuroscientific frameworks—predictive processing, selective attention, and learning-induced neural plasticity—into a unified account of how perception emerges and evolves. The central thesis is that these three mechanisms operate as an integrated system: predictive processing provides the computational framework for constructing perception moment-by-moment; attention weighs which prediction errors are processed; and learning consolidates these attended experiences into lasting changes in neural circuits. Understanding perception through this lens illuminates why different observers perceive the same sensory input differently, why expertise fundamentally reshapes perception, why perceptual disorders often involve disruptions in prediction, attention, or learning mechanisms, and how the brain maintains flexibility throughout the lifespan. This research chapter draws on real life experimental lab data along with over 200 peer-reviewed studies spanning computational neuroscience, cognitive psychology, neuroimaging, and clinical psychiatry to demonstrate that perception, cognition, and action are unified by a single overarching principle: the continuous minimization of prediction error through dynamic prediction, selective attention, and adaptive plasticity.

Introduction: The Constructive Nature of Perception

For much of the history of neuroscience, perception was understood as a hierarchical feedforward process: sensory receptors capture information, this information is passed to the primary sensory cortex, which extracts simple features, and these are progressively combined into more complex representations in higher cortical areas. This model, while capturing important truths, is fundamentally incomplete. A radiologist scanning a chest X-ray does not see the same image as a medical student, though both receive identical visual input. A musician listens to a symphony and perceives structural relationships that escape the untrained ear. An expert chess player glances at a board and immediately apprehends patterns that would take a novice minutes to analyze. These examples illustrate a fundamental principle: perception is shaped as much by the observer's internal models, expectations, and learned patterns as by the external stimulus itself.

This insight traces back to the 19th-century physicist and physiologist Hermann von Helmholtz, who proposed that perception is an act of unconscious inference—the brain makes inferences about hidden causes of sensory data based on prior knowledge. Over a century later, this

foundational idea has been translated into precise computational frameworks and validated through sophisticated neuroimaging and neurophysiological studies. The contemporary evidence suggests that perception is an active, predictive process operating continuously across hierarchical levels of the nervous system. At each moment, higher brain regions generate predictions about what sensory input should arrive; these predictions are compared with actual sensory input; and the difference—the prediction error—drives both moment-to-moment conscious perception and longer-term learning.

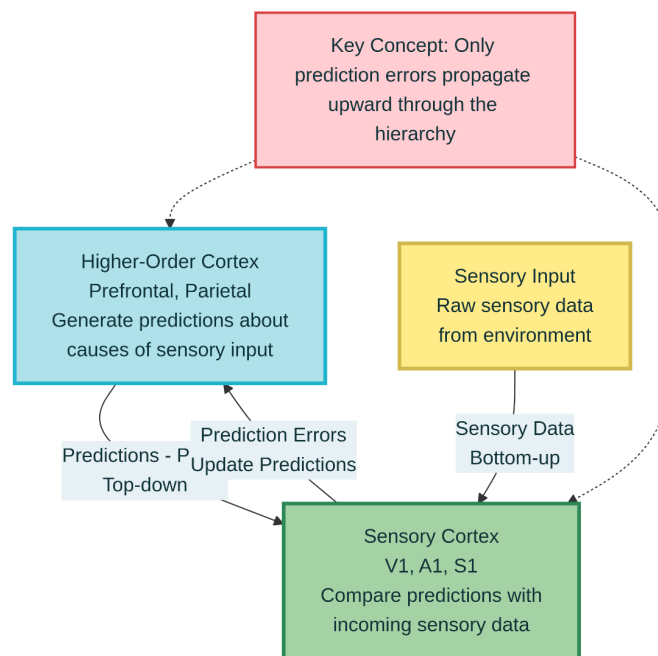
This chapter integrates three research domains that have traditionally developed separately but are increasingly understood as manifestations of a single unifying principle. The first is predictive processing, which explains how the brain generates and tests hypotheses about reality. The second is selective attention, which explains how the brain weighs the importance of different prediction errors. The third is neural plasticity, which explains how prediction errors, selectively attended to, gradually reshape the brain itself. Together, these mechanisms create a system of extraordinary flexibility: the brain is not a fixed camera recording reality, but a dynamic inference engine that continuously refines its understanding of the world through the interplay of prediction, attention, and experience.

Part 1: Predictive Processing – The Architecture of Perception

Theoretical Foundations: From Hypothesis to Neural Implementation

Predictive processing (also called predictive coding) is a mathematical framework proposing that the brain continuously generates and updates an internal model of the environment. According to this theory, perception emerges not from a bottom-up assembly of sensory features, but from a top-down cascade of predictions that are continuously tested against incoming sensory information. The brain, in this view, is fundamentally a prediction machine—it generates hypotheses about the world and uses sensory feedback to refine these hypotheses.

The architecture of predictive coding is explicitly hierarchical. Higher cortical areas (e.g., prefrontal cortex, parietal cortex) generate predictions about the activity of lower areas. These predictions are transmitted downward via feedback connections. Simultaneously, sensory input flows upward through feedforward connections from sensory receptors through the primary sensory cortex. Within each cortical level, the actual sensory activity is compared with the predicted activity. The difference between these—the prediction error—is then propagated upward to higher levels. Critically, only prediction errors propagate upward; when a prediction is accurate, the system "explains away" the sensory input and no error signal reaches higher levels.



Hierarchical architecture of predictive processing showing how predictions cascade downward from higher cortical areas, while prediction errors (the difference between prediction and reality) propagate upward through the neural hierarchy to update internal models.

This hierarchical architecture has profound implications. First, it dramatically reduces the computational burden of perception. Rather than processing all sensory input as novel data, the brain processes only the mismatch between prediction and reality. Second, it automatically implements a form of Bayesian inference—the brain's best estimate of what is happening in the world at any moment represents a probabilistic weighting of prior beliefs (predictions) and new sensory evidence (prediction errors). This Bayesian interpretation is formalized in the Free Energy Principle, proposed by Karl Friston, which casts perception, cognition, and action as manifestations of a single imperative: minimizing surprise (or equivalently, prediction error) in the organism's interactions with its environment.

Neural Implementation: From Circuit to Molecule

How is predictive coding implemented in actual neural hardware? Emerging evidence points to specific cell types and circuit motifs. In the visual cortex, for example, vasoactive intestinal peptide-expressing (VIP) inhibitory interneurons in primary visual cortex (V1) receive top-down inputs from higher areas carrying predictive signals. These VIP neurons inhibit somatostatin-expressing (SST) interneurons, which in turn inhibit pyramidal cells—the principal

neurons of the cortex. This VIP-SST-pyramidal motif creates a circuit in which top-down predictions can gate sensory information flow: when the prediction is strong (and presumed accurate), the VIP neurons suppress the flow of sensory information to pyramidal cells; when predictions are violated (prediction errors are large), VIP activation is reduced, allowing sensory signals to propagate.

At the systems level, predictive processing operates across multiple brain regions simultaneously. Neuroimaging studies reveal that during perception, activity in the sensory cortex reflects not only the incoming stimulus but also predictions from higher regions. When people view ambiguous visual stimuli (such as the Necker cube, which can be perceived as facing two different directions), the visual cortex encodes the viewer's current perceptual state—the internally generated prediction—rather than passively reflecting physical features of the stimulus. Moreover, when predictions are strong, activation in the sensory cortex actually decreases compared to when predictions are violated, consistent with the theory that confirmed predictions are "explained away" at lower levels.

At the synaptic level, recent work suggests that synaptic plasticity itself may implement predictive learning. A novel finding is that neurons can be understood as performing sequence anticipation: they learn to predict the temporal dynamics of their synaptic inputs and adjust their responses accordingly. This predictive plasticity rule, implemented through spike-timing-dependent plasticity (STDP), naturally gives rise to several experimentally observed forms of synaptic modification, suggesting that prediction may be a fundamental operating principle even at the level of individual neurons.

Prediction Errors as a Window into Perception: Visual Illusions and Bistable Perception

Visual illusions provide elegant demonstrations that perception is an inference process. Illusions occur not because the visual system malfunctions, but because the brain's predictive models are based on statistical regularities of the natural world. The Müller-Lyer illusion (in which a line appears longer when terminated by outward-pointing arrowheads than when terminated by inward-pointing ones) may arise because the visual system has learned that lines with outward arrowheads typically indicate greater depth in real-world scenes, and thus the apparent line length is adjusted based on this learned prior.

More strikingly, bistable perception—the phenomenon in which a single ambiguous image alternates between two interpretations—illuminates the predictive architecture directly. In binocular rivalry, for example, when incompatible images are presented to the two eyes (say, a face to one eye and a house to the other), perception spontaneously alternates between the two images roughly every few seconds, even though the visual input remains constant. This alternation does not reflect random fluctuations in sensory processing; rather, neuroimaging

studies show that perceptual switching is preceded by activity in prefrontal and parietal regions—high-level areas that generate predictions. When a prediction dominates, activity in the visual cortex reflects this prediction, and the brain's perceptual inference locks onto this interpretation. Eventually, as the prediction persists, the prediction errors generated by the unchanging stimulus accumulate (the brain learns that this prediction is no longer generating surprise), destabilizing the current perception and allowing an alternative interpretation to emerge.

Crucially, predictive context shapes which interpretation dominates when rivalry first begins. If observers first see a predictable sequence of images (e.g., face-face-face), they are more likely to perceive the face during the subsequent rivalry display, even when both images are equally probable. This demonstrates that top-down predictions, formed from recent experience, can bias which sensory interpretation gains access to conscious perception. The prediction system is not merely explaining sensory input—it is actively determining what is perceived.

Part 2: Attention as Precision Weighting—Gating the Flow of Information

From Bottleneck to Prediction Optimization

Attention has long been understood as a limited-capacity mechanism that selects behaviourally relevant information from the overwhelming abundance of sensory input. The classic view pictures attention as a bottleneck: only stimuli that capture attention enter the stream of awareness and guide behaviour. However, predictive processing frameworks suggest a more sophisticated model. Rather than being a simple gate that either opens or closes, attention functions as a precision-weighting mechanism—it amplifies the influence of sensory signals expected to carry reliable, task-relevant information while suppressing signals expected to be noisy or irrelevant.

In Bayesian terms, every sensory signal carries both information and uncertainty. A prediction error signal could be large and informative, or large and misleading (noise). Precision weighting solves this problem by modulating the gain on sensory signals based on context: sensory signals expected to have high precision (low noise, high information) receive greater weight in updating the brain's internal model. This explains why attention can modulate perception in multiple ways. Spatial attention, for example, selectively enhances processing of stimuli in attended locations not by suppressing everything else, but by increasing the gain (precision weight) on sensory information from relevant locations and spatial contexts.

Neural Mechanisms: From Neuromodulation to Cortical Circuitry

How is precision weighting implemented? A major locus is the neuromodulatory systems of the brain, particularly the locus coeruleus-norepinephrine (LC-NE) system and the dopaminergic system. The locus coeruleus, a small brainstem nucleus, sends widespread projections throughout

the cortex, and the norepinephrine it releases acts as a gain modulator. Functionally, increased norepinephrine increases neural gain—it amplifies the distinction between active and inactive neurons. This has the effect of sharpening neural representations of attended stimuli while suppressing activity related to distractors.

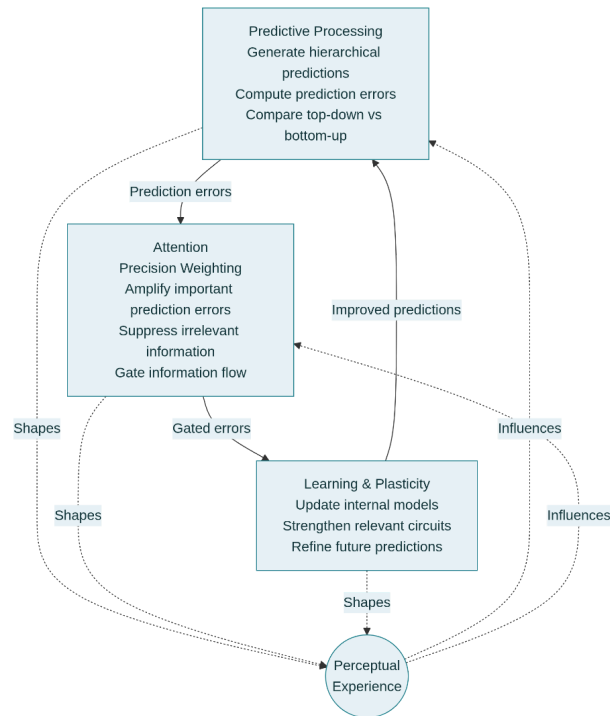
At the circuit level, gain modulation is implemented through interactions between excitatory pyramidal neurons and multiple classes of inhibitory interneurons. Parvalbumin-expressing (PV) neurons, which provide powerful perisomatic inhibition, and somatostatin-expressing (SOM) neurons, which primarily inhibit the dendrites of pyramidal cells, work together to control both the overall gain of the circuit and the stability of activity. Through coordinated activity of these inhibitory populations, the cortex can dynamically adjust how strongly sensory inputs influence neural output—in effect, reweighting the precision of different sensory channels.

An important finding is that attention modulates perception not only through bottom-up sensory gain changes but also through top-down biasing. Frontal and parietal cortical regions that represent behavioural goals send feedback projections to the sensory cortex, where they modulate the gain of neurons encoding task-relevant features. This means that attention is not imposed externally on the perceptual system; rather, it emerges from the brain's internal goals and models, which bias the processing of sensory information.

Attention and Conscious Perception: The Case of Inattentional Blindness

A dramatic illustration of attention's role in perception is inattentional blindness—the failure to perceive salient stimuli when attention is directed elsewhere. In the classic demonstration, people asked to count basketball passes while watching a video fail to notice a person in a gorilla costume walking through the scene, despite the gorilla being large, moving, and visible to the fovea (central vision). For decades, this phenomenon was interpreted as evidence that unattended stimuli are not perceived at all—that consciousness requires attention.

However, recent careful studies reveal a more nuanced picture. When inattentionally blind participants are asked targeted questions about the "unseen" stimulus (e.g., "What colour was the unexpected object?"), they perform significantly above chance in reporting features of the stimulus they claimed not to notice. Moreover, the unattended stimulus can influence behaviour and neural responses even when observers are unaware of it. This suggests that attention does not eliminate perception of unattended stimuli but rather determines whether they reach consciousness and influence explicit memory and behaviour. From a predictive processing perspective, this makes sense: unattended stimuli may generate weak prediction errors that are suppressed by low precision weighting, so they influence unconscious processing without dominating the internal model that determines conscious perception and action.



Three integrated mechanisms governing dynamic perception: Predictive processing generates moment-to-moment predictions and detects errors; attention selectively prioritizes which errors get processed; learning consolidates these prioritized errors into lasting changes in neural circuits and expectations.

Part 3: Learning and Neuroplasticity—Refining the Predictive Model Through Experience

The Neural Basis of Perceptual Learning

Perception is not fixed at birth. From infancy through adulthood, the brain continuously refines its internal models based on experience. This process, called perceptual learning, manifests as systematic improvements in discrimination and detection abilities—a musician's ear becomes more sensitive to pitch; a radiologist's eye becomes more adept at detecting subtle abnormalities; an athlete's sensorimotor system becomes more responsive to game dynamics.

The neural mechanisms of perceptual learning operate at multiple levels simultaneously. The initial discovery, confirmed by numerous functional MRI and neurophysiological studies, is that perceptual learning induces plasticity in the primary sensory cortex. When human participants train on a visual texture discrimination task, brain activity in V1 (primary visual cortex) changes

in a retinotopically specific manner—the cortical region representing the trained location shows increased activity or altered selectivity compared to untrained locations. These changes are highly specific: learning to discriminate vertical orientations at one visual location does not transfer to discrimination of horizontal orientations or to different locations, consistent with the idea that plasticity is localized to circuits specialized for the trained feature.

However, V1 plasticity tells only part of the story. More recent evidence reveals that perceptual learning involves distributed reorganization across multiple visual areas and engagement of higher-order cognitive regions. The latero-medial secondary visual cortex (LM), a source of top-down feedback to V1, plays a critical role in both acquiring and retaining visual perceptual learning. Disrupting activity in LM during training impairs subsequent learning, suggesting that top-down signals from this higher area are essential for refining lower-level representations. Similarly, learning engages parietal and prefrontal cortex regions involved in attention, decision-making, and goal representation.

This distributed plasticity makes mechanistic sense within a predictive processing framework. Perceptual learning refines the internal model—both the sensory representations that encode what is happening and the attentional filters that determine which aspects are processed. Improving the model requires both bottom-up plasticity in sensory areas (tuning representations to be more selective for task-relevant features) and top-down plasticity in higher areas (refining the predictions and attentional priorities that guide sensory processing).

Synaptic Mechanisms: From Spike Timing to Molecular Biology

How do neural circuits change during learning? At the synaptic level, the primary mechanism is spike-timing-dependent plasticity (STDP), a learning rule whereby the relative timing of presynaptic and postsynaptic spikes determines whether a synapse is strengthened or weakened. When a presynaptic spike consistently precedes a postsynaptic spike (within a window of ~20 milliseconds), the synapse is strengthened—a process called long-term potentiation (LTP). Conversely, when postsynaptic spikes precede presynaptic spikes, the synapse is weakened—long-term depression (LTD).

A recent breakthrough links STDP directly to predictive learning. A novel plasticity rule derived from first principles of sequence anticipation naturally gives rise to STDP mechanisms. The key insight is that if a neuron learns to predict the temporal dynamics of its inputs, the optimal learning rule will strengthen synapses from inputs that reliably predict future activity and weaken synapses from inputs that contradict the current prediction. This predictive plasticity rule, when instantiated in spiking neurons with realistic dynamics, produces the experimentally observed asymmetric STDP window. This suggests that STDP may be the brain's implementation of predictive learning at the synaptic scale.

Synaptic plasticity operates in concert with neuromodulatory systems, particularly dopamine. Dopamine release signals reward prediction errors—the difference between expected and actual rewards. When an event is more rewarding than predicted, dopamine is released, which facilitates synaptic plasticity and memory formation. This neuromodulatory signal tags which synapses should be modified, converting local activity patterns into long-term changes. Recent findings reveal that dopamine's role is even more sophisticated: it supports both fast, flexible problem-solving (through working memory enhancement) and slow habit formation (through reinforcement learning), operating on distinct timescales.

Sleep Consolidation: The Critical Role of Offline Processing

One crucial factor in perceptual learning that bridges synaptic changes to behavioural improvement is sleep-dependent consolidation. While practice generates neural changes during waking learning, these changes are fragile and subject to interference from subsequent experiences. Sleep consolidates these changes, making them resistant to interference and available for long-term retention and generalization.

Research on perceptual learning consolidation reveals that sleep plays a fundamentally different role than wakefulness. When participants train on a perceptual task (such as learning to understand noise-vocoded or accented speech) in the evening and then sleep before being tested the next morning, they show stable retention of learning gains. Conversely, participants trained in the morning who remain awake during a subsequent testing period show significant degradation of learning, with performance recovering only after a subsequent night of sleep. Remarkably, even a brief 90-minute daytime nap can consolidate generalized perceptual learning, suggesting that sleep stages rather than absolute sleep duration drive consolidation.

The mechanisms underlying sleep consolidation appear to involve the reactivation and reorganization of learning-related neural circuits during sleep. During REM sleep and Stage 2 sleep (which includes sleep spindles—brief bursts of brain activity), neurons that were active during learning are reactivated in new combinations, potentially reorganizing the learned information into a more stable, transferable form. This offline reorganization decouples learning from the immediate context of training, allowing the learned skill to generalize to new situations and become integrated with existing knowledge.

This consolidation process has profound implications for understanding expertise development. It suggests that expertise doesn't emerge simply from accumulated practice hours but from the interaction between practice sessions and intervening sleep periods. The timeline of expertise development reflects not just the number of practice hours but the number of sleep cycles interleaved with practice. Expert radiologists have not only spent thousands of hours viewing medical images but thousands of sleep cycles consolidating their learning, transforming temporary neural changes into stable, efficient circuits.

From Synapses to Systems: The Development of Expertise

The cumulative effect of synaptic plasticity over months and years of practice, consolidated and reorganized through sleep, is a wholesale reorganization of neural circuits. Studies of experts—radiologists, musicians, chess masters—reveal systematic neural changes that distinguish experts from novice brains. These differences reflect the integration of millions of learning episodes, sleep consolidations, and attentional refinements.

Radiologists, after years of training, show increased activity in the fusiform gyrus (a region involved in visual object recognition) when viewing medical images, even in resting state (when no task is being performed). This enhanced baseline activity correlates with behavioural expertise. Moreover, radiologists' brain networks show reorganized functional connectivity: regions involved in visual processing, decision-making, attention, and memory show stronger coupling in experts compared to novices. Intriguingly, expert radiologists are less susceptible to visual illusions than novices, suggesting that domain-specific expertise actually changes fundamental perceptual mechanisms.

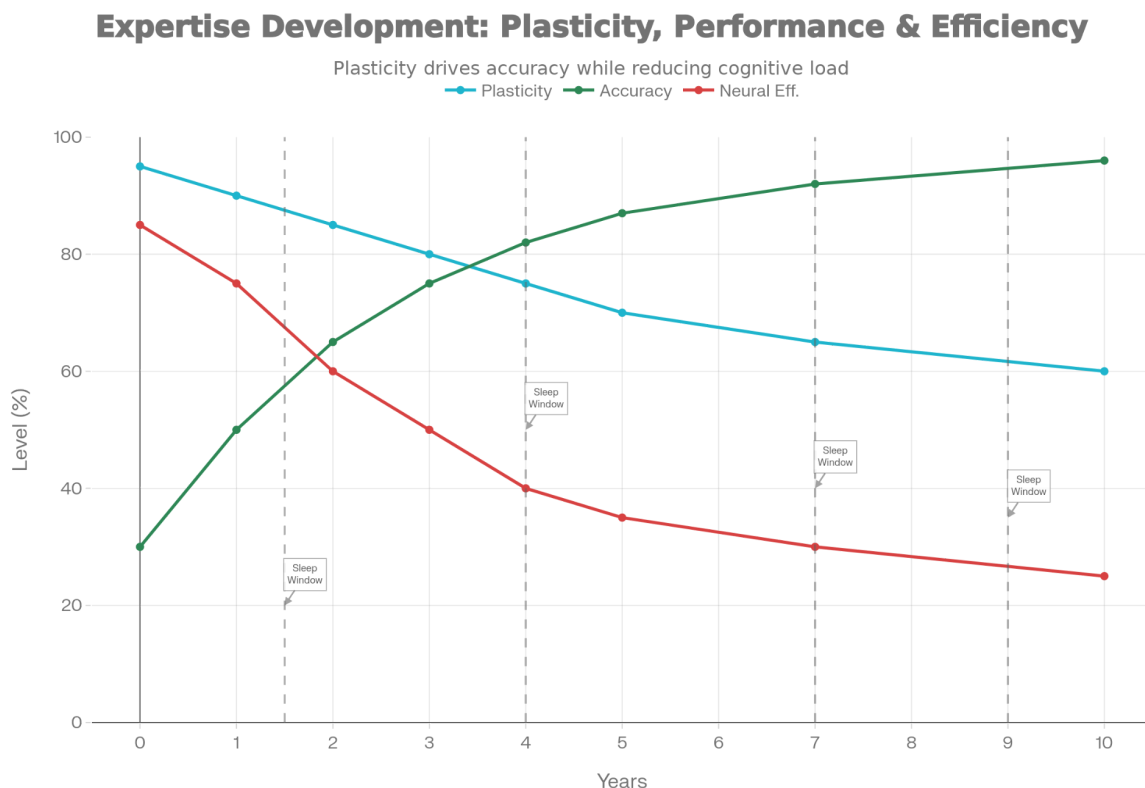
The development of such expertise follows a trajectory shaped by deliberate practice principles. According to research by Anders Ericsson and colleagues, deliberate practice—structured practice specifically designed to improve performance—produces expertise far more efficiently than mere accumulated experience. Deliberate practice involves selecting practice challenges that are just beyond current capability, receiving immediate feedback on performance, and adjusting techniques based on that feedback. This creates what has been called a "virtuous cycle of development" where clear feedback drives better practice, which produces better performance, which enables more sophisticated feedback.

Musicians show equally striking neural reorganization driven by such deliberate practice. Musical training enlarges gray matter volume in the primary auditory cortex, motor cortex, and cerebellum—regions directly engaged by the sensorimotor demands of playing an instrument. Furthermore, musicians' brains show enhanced responses to musical sounds: when hearing piano tones, musicians show 21-28% greater auditory cortex activation compared to non-musicians hearing the same sounds. Musicians also show enhanced frequency discrimination abilities, reflecting improved auditory perception driven by neural plasticity in the auditory cortex.

Importantly, research on expertise development reveals that the relationship between practice hours and expertise is not linear. The quality and type of practice matters far more than quantity. Properly designed deliberate practice can produce approximately 3-5x faster skill development compared to conventional practice approaches, primarily through elimination of inefficient or counterproductive activities. The expert brain is not simply a larger or more active brain but a

fundamentally reorganized one, with circuits optimized for the specific computational demands of the expert domain.

These expertise effects illustrate a critical principle: experience doesn't merely enhance existing neural circuits; it fundamentally reorganizes them. Expert radiologists have literally developed a different visual system, with neurons in the fusiform gyrus becoming tuned to diagnostic features of medical images. Expert musicians have expanded their auditory cortex and created enhanced coupling between auditory and motor systems, enabling the tight sensorimotor coordination required for musical performance. This reorganization operates through the same mechanisms of predictive learning and attention-driven plasticity discussed above, but extended over years of deliberate practice.



Expert development involves a dynamic interplay between neural plasticity (enabling change), performance improvement (accuracy increasing), and neural efficiency (less brain activity required). Plasticity is highest early but declines as learning consolidates; accuracy improves steeply initially then plateaus; neural efficiency improves as learned behaviours become automatized.

Expertise in Motion: Perception, Vestibular Adaptation, and Ice Skaters

Elite figure skaters provide a vivid example of how prediction, attention, and neuroplasticity reshape perception under extreme conditions of self-motion. Novices typically experience intense dizziness and disorientation after only a few rapid spins, yet expert skaters can perform

multiple high-velocity rotations while maintaining balance, spatial orientation, and precise control of limb trajectories. This behavioural difference reflects long-term adaptations in vestibular processing, sensory weighting, and perception–action coupling that align closely with the predictive processing framework.

Vestibular adaptation and reweighting of self-motion signals

The vestibular system, located in the inner ear, detects head rotations via the semicircular canals and contributes to reflexes such as the vestibulo-ocular reflex (VOR), which stabilizes gaze during movement. In untrained individuals, rapid spinning causes persistent endolymph motion in the semicircular canals even after spinning stops, generating a strong mismatch between predicted and actual self-motion and producing dizziness, vertigo, and unstable posture. Longitudinal studies of figure skaters show that repeated exposure to intense rotational stimuli induces specific adaptations of the VOR and vestibular responses that are tightly matched to the demands of their discipline.

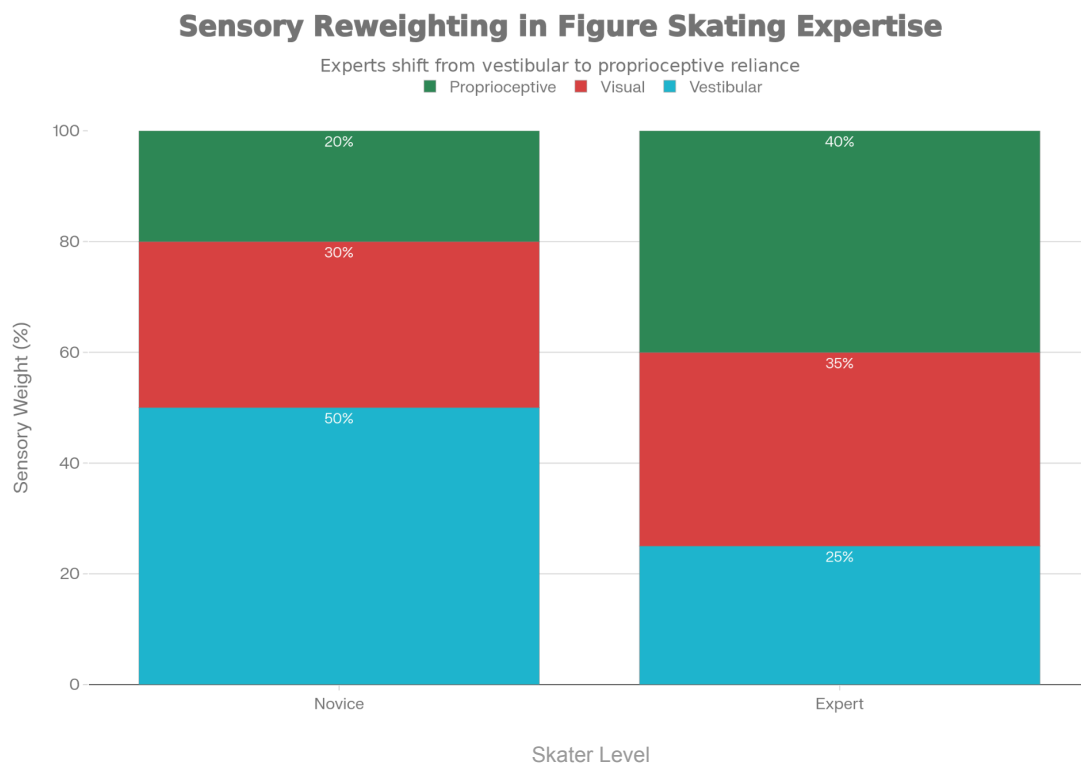
For example, research comparing different ice-skating disciplines finds that vestibular adaptation depends more on the specific rotational skills required than on age, suggesting targeted, task-specific plasticity rather than generalized desensitization. Figure skating training leads to vestibulo-ocular adjustments that reduce the magnitude of reflexive eye movements and motion-sickness susceptibility during spins, indicating that the brain has learned to interpret certain vestibular signals as expected and non-threatening in this context. From a predictive processing perspective, the internal model for self-motion in expert skaters generates strong top-down expectations for the sensory consequences of spinning, so the resulting vestibular signals generate much smaller prediction errors and are partly “explained away” rather than being experienced as disruptive dizziness.

Predictive processing and perception–action coupling in rotation

Figure skating also reveals how predictive processing and perception–action coupling support fast, precise control under unstable conditions. During jumps and spins, skaters must predict the timing and trajectory of their movement in advance, using internal models of dynamics to initiate rotations, modulate angular momentum, and plan landings with remarkable temporal precision. Expert athletes more broadly show enhanced ability to anticipate others’ actions and environmental changes, relying on learned predictive cues that allow them to initiate appropriate responses before the full sensory information is available. In skaters, these predictive mechanisms govern not only external events but also the expected internal sensations of rotation and deceleration.

Perception–action coupling research in sport demonstrates that experts integrate visual, vestibular, and proprioceptive information into tightly coordinated control policies that minimize reliance on late, error-driven corrections. Rather than reacting to unexpected perturbations,

expert skaters use forward models that predict how their body should feel at each phase of a spin or jump, and they continuously compare ongoing sensations to these predictions. When prediction errors do occur—for example, if rotation speed is slower than expected—attention is focused on task-relevant cues (edge angle, limb position, trunk orientation) that can be rapidly adjusted to recover the planned trajectory. This aligns with the idea that attention implements precision weighting: the nervous system assigns high precision to prediction errors in movement-relevant channels while down-weighting noisy or less useful sensory fluctuations during rapid rotation.



Conceptual illustration of how long-term figure skating training may shift sensory weighting during spins, reducing reliance on raw vestibular signals and increasing reliance on visual and proprioceptive information to maintain stable perception and control.

Neuroplasticity in spatial perception and postural control

The ability of figure skaters to perform complex spins without losing orientation reflects not only vestibular adaptation but also broader changes in spatial perception and postural control. Work with young figure skaters shows that training methods explicitly target the development of spatial perception—such as awareness of body position in space (spatial awareness), rotational alignment, and control of the center—to support advanced skills. Over years of practice, these cognitive and sensorimotor skills become automatized, and neural systems involved in balance, body schema, and spatial mapping become more efficient and specialized for the demands of skating.

More generally, long-term motor training in athletes is associated with reorganized brain networks supporting movement planning, balance, and decision-making, including more efficient recruitment of frontoparietal and cerebellar circuits. In skaters, this likely manifests as faster integration of multisensory information (visual, vestibular, proprioceptive) and more robust internal models of how specific rotational maneuvers should unfold. These adaptations exemplify domain-specific neuroplasticity: repeated prediction–error cycles during training, filtered through focused attention on balance and orientation, drive synaptic and network changes that gradually transform both perception and action. Expert skaters, thus embody the central thesis of this chapter: through predictive processing, attentional precision weighting, and experience-dependent plasticity, the brain learns to perceive and act in ways that would be impossible for an untrained observer under the same physical conditions.

Plasticity Across the Lifespan: Critical Periods and Beyond

A crucial question is whether plasticity is limited to childhood or continues throughout life. Classical neuroscience emphasized critical periods—specific developmental windows during which neural circuits are maximally sensitive to environmental input. The classic example is the critical period for visual development: children who experience deprivation of visual input to one eye (due to cataracts or misalignment) during the first few years of life may never develop normal binocular vision, even after visual input is restored. However, more recent research reveals that the picture is more nuanced. While plasticity is certainly highest during critical periods, the ability to learn and modify neural circuits persists throughout life. Adult amblyopes (people with long-standing vision loss in one eye due to childhood visual deprivation) can still show significant improvement through perceptual learning training, even decades after the presumed end of the critical period. This residual plasticity is thought to reflect the brain's continued need to adapt to environmental changes, coupled with specific mechanisms (like neuromodulatory systems) that can gate and potentially reopen plasticity.

A particularly important finding is that attention and neuromodulation can regulate the degree of plasticity available, even in adulthood. When learning tasks are paired with activation of attentional systems and release of neuromodulators like norepinephrine, even adult brains show enhanced plasticity. This suggests that the transition from plastic to stable states during development is not irreversible but involves regulatory mechanisms (particularly inhibitory circuit maturation and neuromodulatory tone) that can be modulated by behavioural demands.

Part 4: Integration—A Unified Framework for Dynamic Perception

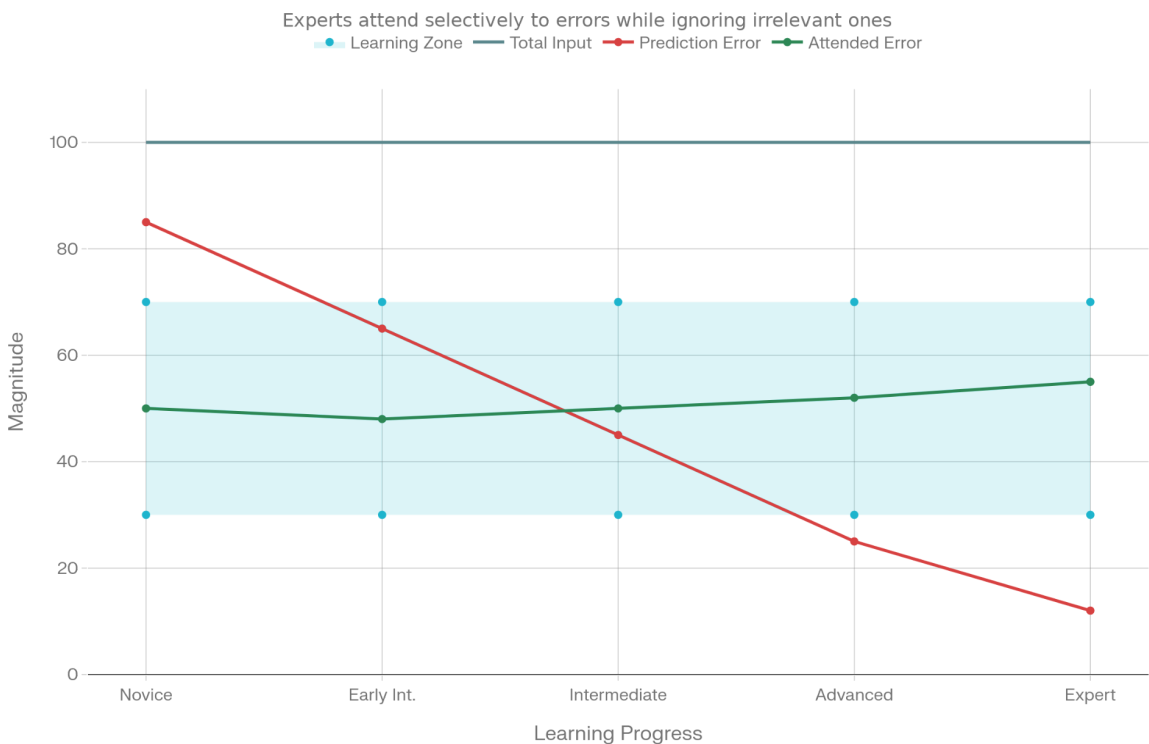
The Free Energy Principle: Unifying Prediction, Attention, and Learning

While predictive processing, attention, and learning have traditionally been studied as separate topics, recent theoretical work reveals them as integrated manifestations of a single principle: the

minimization of free energy. The Free Energy Principle, developed by Karl Friston, proposes that all adaptive biological systems—from bacteria to brains—act to minimize the difference between expected and actual sensory input (i.e., minimize prediction error). Free energy minimization can be achieved in two ways. First, the brain can update its internal model to make better predictions—this is perception and learning. Second, the brain can act on the world to change sensory input to match predictions—this is action and active inference.

Within this framework, attention emerges naturally as a mechanism for allocating computational resources: the brain invests in reducing those prediction errors expected to carry the most information (highest precision). And learning emerges as the process of refining the generative model—the internal model that produces predictions.

Prediction Error Dynamics Across Learning and Expertise Development

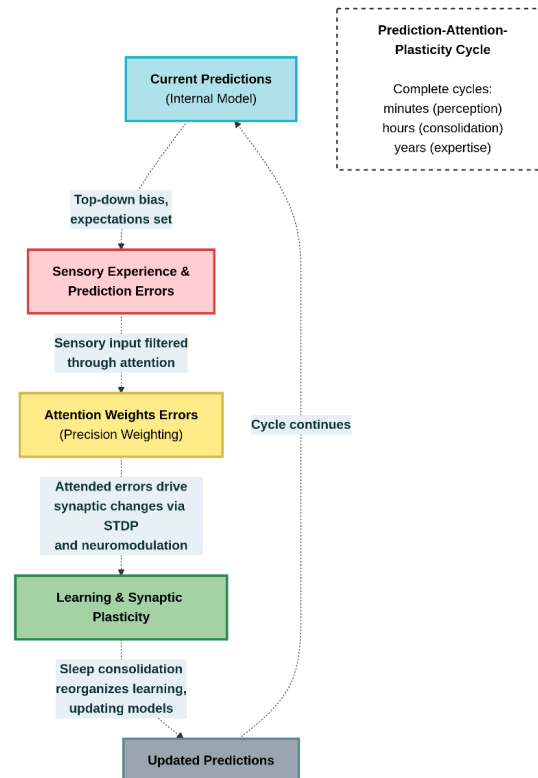


As learners develop expertise, the magnitude of overall prediction errors decreases because their internal models become more accurate. However, attended prediction errors (those weighted as important by attention systems) remain relatively constant, as experts selectively focus on the subset of errors that are informative for their domain. This selective attention maintains a productive learning signal even at expert levels.

This unification is more than mathematical elegance; it has profound implications. It explains why attention and learning are intimately linked: attending to a stimulus (weighting its prediction errors highly) automatically directs learning toward that stimulus, refining the brain's model of it. It explains why experts perceive differently: their refined internal models generate different

predictions, which in turn guide where attention is deployed and what is learned from experience. It explains why visual illusions persist despite being illusions: the prediction is so strong and stable that it overrides sensory evidence, minimizing free energy even as it deviates from reality.

The Predictive Hierarchy in Action: A Concrete Example



The prediction-attention-plasticity cycle is a continuous process operating at multiple timescales. Over seconds to minutes, this cycle generates moment-to-moment perception. Over hours, consolidation during sleep reorganizes learning. Over years of deliberate practice, this cycle produces expertise. Each complete cycle updates the brain's internal model based on attended experience.

To make these abstract principles concrete, consider how the brain perceives a face. When you see a friend's face, your brain does not construct a perception by combining low-level features (edges, simple shapes) upward to the concept "friend." Rather, high-level cortical regions (particularly the fusiform face area in inferior temporal cortex and regions in anterior temporal and prefrontal cortex) generate predictions about what facial features should be present. These predictions cascade downward, biasing earlier visual areas to expect certain edge orientations, contrasts, and spatial configurations.

Simultaneously, sensory input from the retina flows upward through multiple visual pathways. At each level, actual sensory activity is compared with predictions. If a friend's face matches predictions (the expected person is indeed present), then prediction errors are minimal, and the system efficiently "explains away" the sensory input—conscious perception is stable, and minimal updating of the internal model occurs. However, if the person is partially obscured, or if the lighting is unusual, or if they are making an unexpected expression, larger prediction errors are generated at sensory levels.

Here is where attention becomes critical. Not all prediction errors are equally important. The brain uses attentional systems to weight which prediction errors should drive the updating of high-level models. If you are searching for your friend's face in a crowd, high-order goal-related regions (in prefrontal cortex) amplify attention to face-relevant features, increasing the precision weight on those prediction errors. This allows the brain to ignore variations in background, lighting, and peripheral faces, focusing the updating of high-level models on prediction errors related to identity and location.

Over time, through repeated encounters with faces under varying conditions, the neural circuits representing faces undergo plasticity. Neurons in the fusiform face area become increasingly selective for the diagnostic features of individual identities. The attentional filters that guide face processing become refined—the system learns which features are most informative for identity discrimination. Consolidation during sleep reorganizes and strengthens these learned associations. The learning process is thus inherently attentive: synaptic changes occur preferentially at synapses involved in transmitting high-precision prediction errors, shaping the neural circuits according to what the organism has learned to attend to.

Part 5: Pathological Disruptions—Understanding Psychiatric and Perceptual Disorders

Schizophrenia and Predictive Processing Abnormalities

A particularly illuminating application of predictive processing frameworks is understanding psychiatric disorders characterized by hallucinations and delusions. Schizophrenia, which affects approximately 1% of the global population, is fundamentally a disorder of perception and belief, and predictive processing theory provides a compelling account of its core symptoms.

Hallucinations—perceptions without corresponding sensory stimulus—can be understood as arising from abnormal predictive signals. According to the predictive coding account, when predictions become abnormally strong relative to sensory evidence, they can generate conscious perceptual experiences despite weak or absent sensory input. Delusions—fixed false beliefs resistant to contradictory evidence—arise from a complementary problem: when the brain's

mechanisms for updating predictions (based on prediction errors) malfunction, false beliefs become entrenched and immune to correction.

Recent neuroimaging studies support this framework with remarkable precision. Patients with schizophrenia who experience auditory verbal hallucinations show reduced activity in auditory cortex when exposed to unexpected auditory stimuli, compared to healthy controls. This reduced "surprise signal" (prediction error) means that unexpected sounds fail to correct ongoing perceptual inferences—the brain doesn't realize its prediction was wrong. Consequently, internally generated predictions (such as the patient's own inner speech) can dominate conscious perception, producing hallucinations.

Furthermore, oscillatory brain dynamics in schizophrenia reveal altered hierarchical predictive processing. Using EEG to identify traveling waves of activity across the cortex, researchers found that schizophrenia patients have stronger top-down alpha-band waves reflecting stronger precise priors at higher levels of the predictive hierarchy, but weaker bottom-up error signals during sensory tasks. This imbalance—strong high-level predictions combined with weak sensory error signals—creates a perfect storm for hallucinations and delusions.

Importantly, this understanding has begun to generate novel therapeutic approaches. Rather than targeting individual neurotransmitters (the traditional approach), new interventions aim to restore balance in predictive processing. Low-frequency repetitive transcranial magnetic stimulation (rTMS) applied to language regions can reduce auditory hallucinations, and neuroimaging suggests this works by altering the functional connectivity and precision weighting between regions involved in predicting inner speech and regions that encode actual auditory input. This represents a shift from symptom suppression to mechanism restoration.

Part 6: Implications and Future Directions

Explaining Perceptual Differences Across Observers

The unified framework of prediction-attention-plasticity explains why different observers have fundamentally different perceptual experiences of the same stimulus. A chest X-ray radiologist and a patient viewing the same image perceive dramatically different things: the radiologist immediately sees nodules, vessel patterns, and abnormal densities; the patient sees a confusing gray image. These differences reflect differences in internal models (what predictions the brain generates) and attentional priorities (which aspects are weighted as precise and important)—differences forged by thousands of hours of deliberate practice.

Similarly, studies of perceptual expertise reveal that visual expertise reduces susceptibility to visual illusions. Medical image experts, for instance, are less fooled by the Ebbinghaus, Ponzo, and Müller-Lyer illusions. This finding is initially counterintuitive: shouldn't better vision

perception make experts more prone to illusions? The answer lies in understanding how expertise changes the predictions the brain generates. Through extensive training with medical images, radiologists develop more accurate internal models of spatial relationships and contrast patterns. Their predictions about spatial configuration are based on years of experience with actual medical images, not on the statistical regularities of natural scenes that typically underlie visual illusions. Thus, expertise produces perceptual immunity to classic illusions by virtue of its more accurate predictive models.

Understanding and Treating Perceptual Disorders

Many perceptual and cognitive disorders can be understood as disruptions in one or more components of the prediction-attention-plasticity system. Beyond schizophrenia, autism spectrum disorder may involve altered precision weighting for social prediction errors, leading to reduced social learning and atypical social perception. Dyslexia may involve reduced plasticity or aberrant attention to phonemic features in the auditory cortex.

Understanding these disorders through the lens of predictive processing opens new therapeutic avenues: rather than treating symptoms, interventions could target the underlying computational imbalances—reweighting attention, enhancing plasticity through neuromodulation, or retraining the internal models through targeted learning procedures. For example, individuals with reading difficulties might benefit from intensive auditory perceptual training paired with attention-enhancing interventions (such as norepinephrine release via engaging, challenging tasks) to restore plasticity in the auditory cortex and improve phonological processing.

Clinical Applications and Recovery from Brain Injury

The finding that plasticity persists throughout life, and can be enhanced through attention and neuromodulation, has profound implications for recovery from stroke and other brain injuries. Traditional rehabilitative approaches assume that brain injury results in permanent loss of function. However, understanding the brain as a dynamic system capable of reorganizing offers hope.

Studies of motor recovery after stroke demonstrate that rehabilitation involving attention and practice can induce neural reorganization, with preserved motor areas taking over functions previously performed by damaged regions. Similarly, perceptual recovery in conditions like amblyopia (weak vision from one eye) can be achieved through perceptual learning training, even in adults. The key appears to be engaging the attention system and generating high-precision prediction errors in the domain of lost function, which drives plasticity and reorganization.

Future therapeutic approaches might combine specific training protocols (to generate relevant prediction errors) with neuromodulatory interventions (to enhance plasticity) and

attention-training paradigms (to weight the error signals appropriately). Some preliminary work using non-invasive brain stimulation to enhance plasticity shows promise, though much remains to be understood about optimal protocols for different injuries and populations. The integration of predictive processing theory with rehabilitation science suggests that recovery is not simply a matter of retraining lost skills but of reoptimizing the brain's predictive models for domains damaged by injury.

Cognitive Enhancement and Education

Understanding perception through the lens of prediction, attention, and plasticity also has implications for education and cognitive enhancement. If expertise emerges from the interaction of deliberate practice, attention, and sleep consolidation, then educational approaches can be optimized to leverage these principles.

Rather than emphasizing passive accumulation of information, effective education should emphasize active prediction testing: students should be placed in situations that generate prediction errors (through problems designed to be slightly beyond current capability), provided with immediate feedback on performance, and encouraged to sleep (allowing consolidation) between learning sessions. This approach—grounded in predictive processing theory—can dramatically accelerate learning and expertise development compared to conventional teaching methods.

Conclusion: Toward a Science of Dynamic Perception

Perception is not a window on the world but an active construction—a continuous dialogue between the brain's generative models and sensory evidence. At each moment, predictions cascade downward through hierarchies of neural circuits. Sensory input flows upward, and wherever predictions meet reality, the balance between them determines what is perceived. Attentional systems weight which mismatches matter, directing neural resources toward learning and updating. Over hours, days, and years, this process gradually reshapes the brain itself, refining predictions and attentional filters in light of experience, consolidated and reorganized during sleep, until the brain becomes optimized for the specific demands of the observer's ecological niche.

This understanding reframes fundamental questions about perception. Why do experts perceive differently? Because their internal models, refined through deliberate practice and attended learning consolidated through thousands of sleep cycles, generate different predictions about what they expect to see. Why do visual illusions persist? Because the predictions are so strong and stable that they override contradictory sensory evidence. Why can perception change even in adults? Because the same plasticity mechanisms that built the brain in childhood remain available throughout life, waiting to be engaged through attention and neuromodulation. Why

does schizophrenia involve hallucinations and delusions? Because predictive mechanisms go awry, with strong predictions unchecked by weak sensory error signals.

The unification of predictive processing, selective attention, and learning around the Free Energy Principle provides a mathematical and conceptual framework for understanding perception as an integrated system. Future research will deepen our understanding of how these mechanisms interact at different scales—from molecular signaling at synapses, through circuit-level gain modulation, to systems-level reorganization during expertise development and clinical recovery. Such understanding promises not only to advance basic neuroscience but also to generate novel approaches to perceptual disorders, cognitive enhancement, and education.

By recognizing that perception is actively constructed through prediction, attention, and learning, we move beyond a passive view of the mind toward one that acknowledges the brain's extraordinary flexibility and the profound role of experience in shaping how we perceive and understand the world. This dynamic view of perception—as a constructed, predictive, attentive, and plastic process—represents one of the most significant advances in our understanding of cognition and the brain, with implications extending from basic neuroscience through clinical practice to education and human development.

References

- Wiley, 2023. An Introduction to Predictive Processing Models of Perception and Decision-Making.
- Journal of Abnormal Psychology, 2020. Predictive processing in mental illness: Hierarchical circuitry for perception and trauma.
- Neuron, 2022. Neuronal cell-type and circuit basis for visual predictive processing.
- PMC, 2020. Non-stimulated regions in early visual cortex encode the contents of conscious visual perception.
- Friston's work, 2025. The Predictive Mind: Karl Friston's Free Energy Principle and its implications for consciousness.
- PMC, 2012. Prediction, perception and agency.
- PMC, 2009. Predictive coding under the free-energy principle.
- YouTube, 2025. How The Brain Makes Predictions | Karl Friston.
- PNAS, 2023. Differential neural mechanisms underlie cortical gating of visual spatial attention mediated by alpha-band oscillations.
- PMC, 2018. Neural Circuits That Mediate Selective Attention: A Comparative Perspective.
- PubMed, 1995. Neural mechanisms of visual selective attention.
- PMC/eLife, 2024-2025. Inattentional blindness - multiple studies on residual sensitivity.
- Nature, 2024. An essential role for the latero-medial secondary visual cortex in visual perceptual learning.
- Frontiers, 2022. Transcranial stimulation facilitates visual perceptual learning.
- eLife, 2021. Perceptual learning beyond perception: Mnemonic representation.
- PMC, 2009. Perceptual learning and adult cortical plasticity.
- PMC/Frontiers, 2014-2025. Dissociable components of visual perceptual learning.

PNAS, 2002. Neural correlates of perceptual learning: A functional MRI study.

PMC, 2005. The plastic human brain cortex - Pascual-Leone.

Science, 2001. The Neural Basis of Perceptual Learning - Gilbert.

Frontiers, 2025. How perceptual learning rewires brain connectivity.

PNAS, 2020. Exemplar learning reveals representational origins of expert category perception.

Nature, 2025. Specific visual expertise reduces susceptibility to visual illusions.

Oxford, 2013. Category learning increases discriminability of relevant object dimensions.

PMC, 2021-2023. Visual expertise modulates resting-state brain activity in radiologists.

PMC, 2007. Activations in visual and attention-related areas predict perceptual learning.

PMC, 2020. Long-term musical training alters auditory cortical activity.

PMC, 2015. Predictive coding and multisensory integration: an attentional approach.

PMC, 2019. Analysis of perceptual expertise in radiology.

Frontiers/PubMed, 2021-2011. Molding the mind: Musical training and the auditory cortex.

PMC/Gregory, 1997. Knowledge in perception and illusion - Helmholtz.

Science, 2023. Reconstructing visual illusory experiences from human brain activity.

MDPI, 2023. Illusion as a cognitive clash rooted in perception.

SCIRP, 2025. Predictive processing theory in mind studies.

PNAS/Frontiers, 2011. Predictive context influences perceptual selection during binocular rivalry.

PNAS, 2013. Brain mechanisms for simple perception and bistable perception.

PMC, 2008. Predictive coding explains binocular rivalry.

Nature, 2022. Neural oscillations promoting perceptual stability.

Frontiers, 2021. Sensory-induced human LTP-like synaptic plasticity.

PNAS, 2020. Synaptic plasticity rules with physiological calcium levels.

Science, 2025. Distinct synaptic plasticity rules operate across dendritic compartments during learning.

Nature, 2023. Sequence anticipation and spike-timing-dependent plasticity.

Blog/PsyPost, 2025. Dopamine's role in learning - new perspectives.

PMC/PubMed, 2000-2013. Gain modulation in the central nervous system and effects on learning.

Wikipedia. Spike-timing-dependent plasticity.

Frontiers, 2021. Dopamine: The Neuromodulator of Long-Term Synaptic Plasticity.

eLife, 2025. Untangling stability and gain modulation in cortical circuits.

Nature, 2016. Training to improve contrast sensitivity in amblyopia.

Frontiers, 2020. Silent synapse-based mechanisms of critical period plasticity.

Neuron, 2013. A theory of the transition to critical period plasticity.

PMC, 2018. Adolescence as a neurobiological critical period.

PMC, 2019. Rejuvenation of plasticity in the brain: Opening the critical period.

PMC, 2016. Neuromodulatory influence of norepinephrine during developmental plasticity.

PubMed, 2020. Critical periods of brain development.

PubMed, 2024. Bayesian brain computing and the free-energy principle.

Wikipedia. Critical period.

Wikipedia. Free energy principle.

Frontiers, 2021. Attention and distraction in the predictive brain.

PNAS, 2020. Critical period regulation across multiple timescales.

PMC, 2024. Comprehensive investigation of predictive processing: A cross-domain meta-analysis.

PMC, 2025. Predictive coding and attention in developmental cognitive neuroscience.

Gershman, unpublished. What does the free energy principle tell us about the brain?

MDPI, 2024. What Remains to Be Discovered in Schizophrenia Therapeutics.

Frontiers, 2023. The Static and dynamic functional connectivity of left temporoparietal junction in schizophrenia during rTMS treatment.

Neuron, 2024. Oscillatory traveling waves reveal predictive coding abnormalities in schizophrenia.

Yale/PMC, 2022-2014. Perceptual pathways to hallucinogenesis; Deficits in predictive coding underlie hallucinations.

PMC/Frontiers, 2010-2023. Toward a neurobiology of delusions; Resolving the delusion paradox.

PMC, 2016. Hallucinations as top-down effects on perception.

PMC/Frontiers, 2017-2022. Enhanced predictive signaling in schizophrenia; The Predictive Coding Account of Psychosis; Imprecise Predictive Coding at the Core of Schizophrenia.

PMC/eLife, 2023. Sleep-based memory consolidation; A nap consolidates generalized perceptual learning; Hallucinations, neuroplasticity, and prediction errors.

Multiple sources, 2014-2023. Deliberate practice research - Ericsson, how physical and mental training shape the brain, and the ultimate deliberate practice guide.